

Day 23 Hybrid Notes

Assumptions, formulas, constraints, and decisions | Cloze + outline + active learning

Name:	Date:	Period:
-------	-------	---------

Learning Targets
I can choose assumptions and formulas for a modeling problem.
I can use geometry measurements to make a decision.
I can explain whether a model is reasonable and what its limits are.

Part 1: Cloze Notes - Essential Ideas
1. A model is a simplified mathematical version of a real _____.
2. An assumption is something we decide to treat as _____.
3. A constraint is a limit or requirement the solution must _____.
4. Covering, wrapping, and painting usually use _____.
5. Filling, holding, and capacity usually use _____.
6. Scale affects length by k , area by k^2 , and volume by _____.
7. A reasonable answer includes a number, a unit, and a _____.

Part 2: Outline Notes - Big Picture A. Modeling starts with understanding the situation before _____. B. The formula depends on what the question is asking you to _____. C. Units help check whether the answer matches the _____. D. A strong model also names limitations and possible _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ —
---	--

Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 23 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. A wall is 24 ft by 9 ft. Paint covers 350 square feet per gallon. How many gallons are needed?
Setup/work: _____
Answer: _____
2. A scale model has volume 20 cubic inches. The actual object has scale factor 6. Find actual volume.
Setup/work: _____
Answer: _____

Part 4: Error Analysis

A student uses floor area to decide how much air a room holds.
What went wrong?

Correct idea:

—

Before You Solve Checklist

- ☐ I labeled the diagram or graph.
- ☐ I chose the correct relationship or formula.
- ☐ I substituted values carefully.
- ☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that _____.

Sentence 2: One thing I need to remember when solving is _____.

Part 5C: Application Problem

A box must hold at least 700 cubic inches. It measures 12 in by 10 in by 6 in. Does it meet the constraint?
Setup: _____
Answer: _____
Sentence answer: _____